

# Jaymin Sheladia

+1 (213) 331-9625 | sheladia@usc.edu |  LinkedIn |  LeetCode |  GitHub

## EDUCATION

### University of Southern California

Master of Science in Computer Science

Los Angeles, USA

January 2026 – December 2027

### Graphic Era Deemed to be University

B.Tech in Computer Science and Engineering

Dehradun, India

October 2021 – June 2025

## TECHNICAL SKILLS

**Languages:** C, C++, Python, Java, JavaScript, HTML, SQL

**Frameworks/Libraries:** Angular, React, Express.js, Spring Boot, Flask, TensorFlow, Keras, OpenCV, Scikit-learn

**Developer Tools:** Git, GitHub Actions, Docker, VS Code, PyCharm, Jupyter Notebook, Postman

**Databases:** MySQL, MongoDB, PostgreSQL

**Concepts:** Data Structures & Algorithms, Distributed Systems, Unix/Linux, RESTful API Design, CI/CD

**Relevant Coursework:** Analysis of Algorithms, Database Systems, Web Technologies, Applied Natural Language Processing, Agentic AI Systems

## WORK EXPERIENCE

### Cognizant Technology Solutions

May 2025 – September 2025

Programmer Analyst Trainee – Software Engineering

Coimbatore, India

- Engineered a full-stack Travel Planner and Booking System, designing **10+** RESTful APIs in Spring Boot and Angular to modernize manual booking workflows and centralize itinerary, booking, and preference management.
- Integrated a Generative AI-based recommendation engine that analyzes **3** key factors – user interests, budget, and trip duration – to surface personalized suggestions that simplified the booking decision process.

### CODTECH IT SOLUTIONS

June 2024 – August 2024

Machine Learning Intern

Remote

- Designed and trained a fraud detection model to flag fraudulent transactions, applying anomaly detection and supervised learning on over **200,000** credit card transactions to reach **98%** accuracy.
- Optimized algorithms and rebuilt data preprocessing pipelines for existing production models, enhancing their accuracy by **12–15%** across multiple live machine learning projects and reporting dashboards.

## PROJECTS

### TechHire: AI-Powered Job Search Platform | GitHub | React, FastAPI, PostgreSQL May 2026 – June 2026

- Built a full-stack job search platform with a React/Vite frontend and FastAPI/PostgreSQL backend to simplify job discovery, implementing filters across **6** dimensions including skills, salary, and visa sponsorship.
- Leveraged the Groq API to auto-generate AI-powered job summaries and built resume upload, text-extraction, and resume-to-job comparison features backed by automated Python scrapers pulling live listings.

### Sequence Alignment: DP vs. Divide & Conquer | GitHub | Python

March 2026 – April 2026

- Implemented and benchmarked the classical Needleman-Wunsch dynamic programming approach against Hirschberg's memory-efficient divide-and-conquer algorithm for large genomic sequence inputs.
- Reduced memory usage from  $O(m \times n)$  to  $O(m + n)$  using Hirschberg's algorithm, cutting peak memory consumption by up to **90%** on large inputs while preserving identical optimal alignments.

### Real-Time Sign Language Detection | GitHub | Python, OpenCV, TensorFlow

July 2025 – August 2025

- Developed a real-time computer vision system using OpenCV and TensorFlow to classify hand gestures into text for real-time gesture translation, achieving **98%** accuracy across a diverse gesture set.
- Trained and fine-tuned a Keras-based deep learning model on a custom gesture dataset, optimizing preprocessing and inference to support smooth, low-latency real-time video translation for end users.

## RESEARCH

### Deep Learning Approach for Malicious URL Detection | Co-author; Published Research Paper

2025

- Compared **4** deep learning architectures (CNN, RNN, LSTM, Bi-LSTM) for malicious URL detection, engineering character-level preprocessing pipelines on the IEEE Dataport dataset for sequential analysis at scale.
- Achieved **99.25%** accuracy with the Bi-LSTM model, outperforming CNN (**86.5%**), LSTM (**63.75%**), and RNN (**53%**) baselines across standard classification and error-rate metrics.